

Orphan-Mesh Region Swap — Change-Control Ledger

Model: dome_stress_relief — transient thermal cavity-radiation deck

PURPOSE

Track every mesh entity, set, surface, cavity-radiation interaction, load/BC, and section affected by swapping a region of the orphan mesh — so nothing silently loses its boundary condition, drops out of a cavity enclosure, or ends up with an unassigned section.

HOW TO USE TABS

1) Dump deleted & new-region labels (Step 11 of the walkthrough). 2) Run inp_impact_report.m on the pre-swap .inp to auto-populate the 'Impacted?' answers. 3) Paste/confirm rows on each tab. 4) Do the Action. 5) Mark Verified = Y. 6) Summary tab must show 0 outstanding before you submit.

Mesh · Sets · Surfaces · Cavity_Radiation · Loads_BCs · Sections · Summary

STATUS CODES

- OK
- PARTIAL
- FULLY_DELETED
- REFERENCES

Entity does not touch the deleted region — no action.
Entity references SOME deleted labels — append the new-region entities so it stays complete.
Every referenced label was deleted — rebuild entirely on the new region (or retire it).
A surface/cavity/section/load points at an impacted entity — redefine / reassign / recompute.

COLOR KEY

- Yellow fill
- Orange fill

Cells you fill in.
Critical — a silent-failure risk if left undone (BC loss, dropped cavity surface, unassigned section).

SILENT-FAILURE REMINDERS

- Cavity radiation
- Temperature BC
- Section
- Numbering

A rebuilt surface used in a *CAVITY DEFINITION requires *RADIATION VIEWFACTORS to be RECOMPUTED for that cavity, and VFTOT re-checked (should be ~1.0 for a closed cavity).
If the new region belongs on OUT or SINK_I, add its nodes to that set or the region gets no prescribed temperature — the run still completes with a wrong field. Watch OP=NEW.
The new-region elset needs its section assigned. If TYPE=ORTHO conductivity, the ORIENTATION must be the CYLINDRICAL system, not global Cartesian.
New region starts at 20001+ (offset 20000) to stay globally unique — the ledger keys on that.

Region	Instance	Entity type	Label range / set	Count	Action	Label check (≥20001 for new)	Verified	Notes
Deleted region	PART-1-1	Elements	DEL_REGION_ELEMS	480	Deleted from orphan mesh	n/a	Y	Dumped via CAE kernel to deleted_elems.txt
New region	NEWPART-1	Elements	NEWREGION (≥20001)	610	Added, tied to exposed faces	Y	Y	Offset +20000 applied

Set name	Type (NSET/ELSET)	Impacted?	Status	Used by (BC/Load/Output/Surface/Section)	Action needed	New-region entities added?	Verified	Notes
OUT	NSET	Y	PARTIAL	*BOUNDARY (temp, DOF 11)	Append new-region outer-surface nodes so the prescribed T still applies	N	N	Silent-failure risk: unconstrained surface
SINK_I	NSET	N	OK	*BOUNDARY (sink)	None	N/A	Y	Not in swapped region
SS	NSET	Y	PARTIAL	Surface HEAD (cavity C15)	Rebuild set on new faces; then rebuild surface + recompute view factors	N	N	Discontinuous ranges — verify visually

Surface name	Type (ELEMENT/NODE)	Built on (elset/elem/nset)	Face id (S#/SPOS/SNEG)	In cavity?	Impacted?	Action	View factors recomputed?	Verified	Notes
HEAD	ELEMENT	head_s, SS	S1	C15	Y	Redefine on new tied-region faces; confirm face id	Y	N	Shell sidedness (SPOS/SNEG) determines radiating face

Cavity (NAME)	Participating surfaces	Impacted surface(s)	Surface property (ϵ)	Action	*RADIATION VIEWFACTORS recomputed?	VFTOT re-checked (~1.0)?	Verified	Notes
C15	SS_I2, SSO1, SS_B1, SST3	SS_I2	metalF (0.45–0.50)	Rebuild SS_I2 on new region, then recompute view factors for C15	N	N	N	Check enclosure still opaque-bounded after swap

Name / keyword	Type	Applied to (set/surface)	DOF / mag / amplitude	Impacted?	Action	Re-applied to new region?	Verified	Notes
*BOUNDARY	Prescribed temp	OUT	11,11,1.0 (AMP=SR)	Y	Add new outer nodes to OUT; use OP=NEW once, OP=MOD after	N	N	Two OP=NEW cards wipe each other — verify
*SRADIATE / cavity	Cavity radiation	surfaces via cavities	ϵ from metalF	Y	Handled on Cavity_Radiation tab	N	N	

Section keyword	Elset	Material	Orientation (cyl. if ORTHO)	Impacted?	Action	Assigned to new region?	Verified	Notes
*SOLID SECTION	FORGING	STEEL (therm.mat)	CYL-1 (cylindrical)	Y	Assign same section to new-region elset; keep ORIENTATION cylindrical	N	N	TYPE=ORTHO needs *ORIENTATION or conductivity applies on global axes

Change-Control Summary

Outstanding must be 0 before submitting. Counts ignore the grey EXAMPLE rows only if you delete them.

Category	Impacted (Y)	Verified (Y)	Outstanding
Sets impacted	2	1	1
Surfaces impacted	1	0	1
Cavities impacted	1	0	1
Loads/BCs impacted	2	0	2
Sections impacted	1	0	1

TOTAL OUTSTANDING

6